

Q What is the placenta and how does it work?

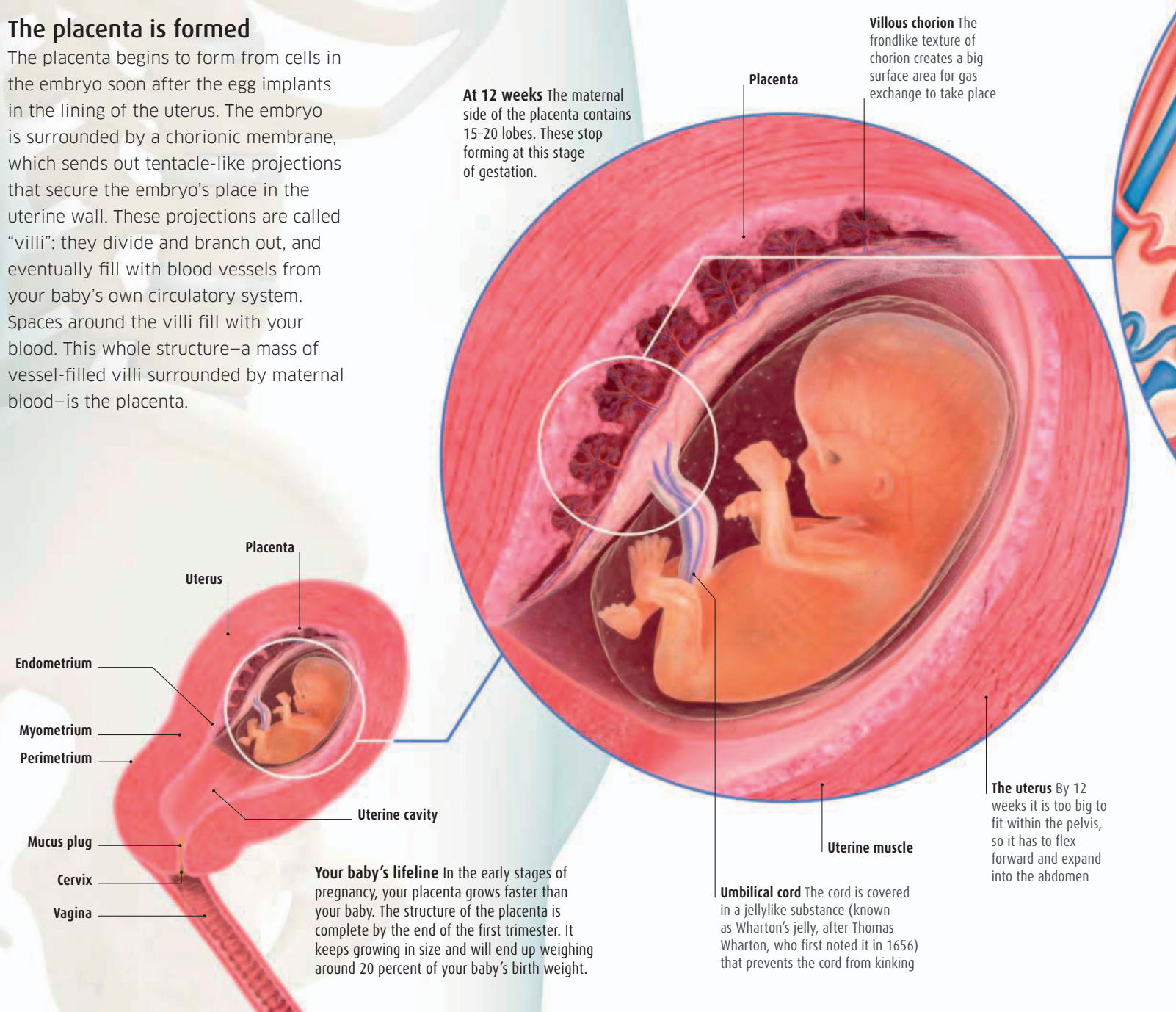
The placenta is your baby's life-support system. It's the means by which she receives oxygen and nutrients, it provides her with protection against disease, and it's where the waste she produces goes to be disposed of. The placenta is part of you and part of her at the same time. It's the link that enables you to nourish her while she grows inside you.

The placenta is formed

The placenta begins to form from cells in the embryo soon after the egg implants in the lining of the uterus. The embryo is surrounded by a chorionic membrane, which sends out tentacle-like projections that secure the embryo's place in the uterine wall. These projections are called "villi": they divide and branch out, and eventually fill with blood vessels from your baby's own circulatory system. Spaces around the villi fill with your blood. This whole structure—a mass of vessel-filled villi surrounded by maternal blood—is the placenta.

At 12 weeks The maternal side of the placenta contains 15–20 lobes. These stop forming at this stage of gestation.

Villous chorion The frondlike texture of chorion creates a big surface area for gas exchange to take place



Your baby's lifeline In the early stages of pregnancy, your placenta grows faster than your baby. The structure of the placenta is complete by the end of the first trimester. It keeps growing in size and will end up weighing around 20 percent of your baby's birth weight.

Umbilical cord The cord is covered in a jellylike substance (known as Wharton's jelly, after Thomas Wharton, who first noted it in 1656) that prevents the cord from kinking

The uterus By 12 weeks it is too big to fit within the pelvis, so it has to flex forward and expand into the abdomen